RQ1 | Test utility across all methods

	AppWorld			PlanBench-XL		
Method	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B
Frozen Base	0.2583	0.2720	0.3341	0.2633	0.2682	0.3290
AREW / T3	0.2771	0.2826	0.3665	0.2809	0.3068	0.3744
GiGPO	0.3094	0.3295	0.3952	0.3163	0.3419	0.4070
InfoPO	0.3634	0.3955	0.5268	0.3975	0.4163	0.5279
WorldEvolver	0.3636	0.3827	0.4487	0.4441	0.4541	0.5227
PoS / EnvProbe	0.5112	0.5583	0.6649	0.4921	0.5627	0.6630
Step-DPO	0.5583	0.5871	0.6770	0.5356	0.5607	0.6643
Outcome-PG	0.2999	0.3138	0.4078	0.3186	0.3520	0.4099
Outcome-Value	0.3084	0.3310	0.4104	0.3316	0.3693	0.4099
Outcome-RLOO	0.3056	0.3188	0.4038	0.3212	0.3546	0.4099
Info-RL Neighbor	0.3350	0.3633	0.4635	0.3693	0.3900	0.4233
Unstratified	0.3444	0.3808	0.4712	0.3926	0.3998	0.4743
Unweighted	0.5842	0.5279	0.6484	0.5932	0.5939	0.6386
SPAR	**0.6130**	**0.6279**	**0.6948**	**0.6243**	**0.6677**	**0.6847**

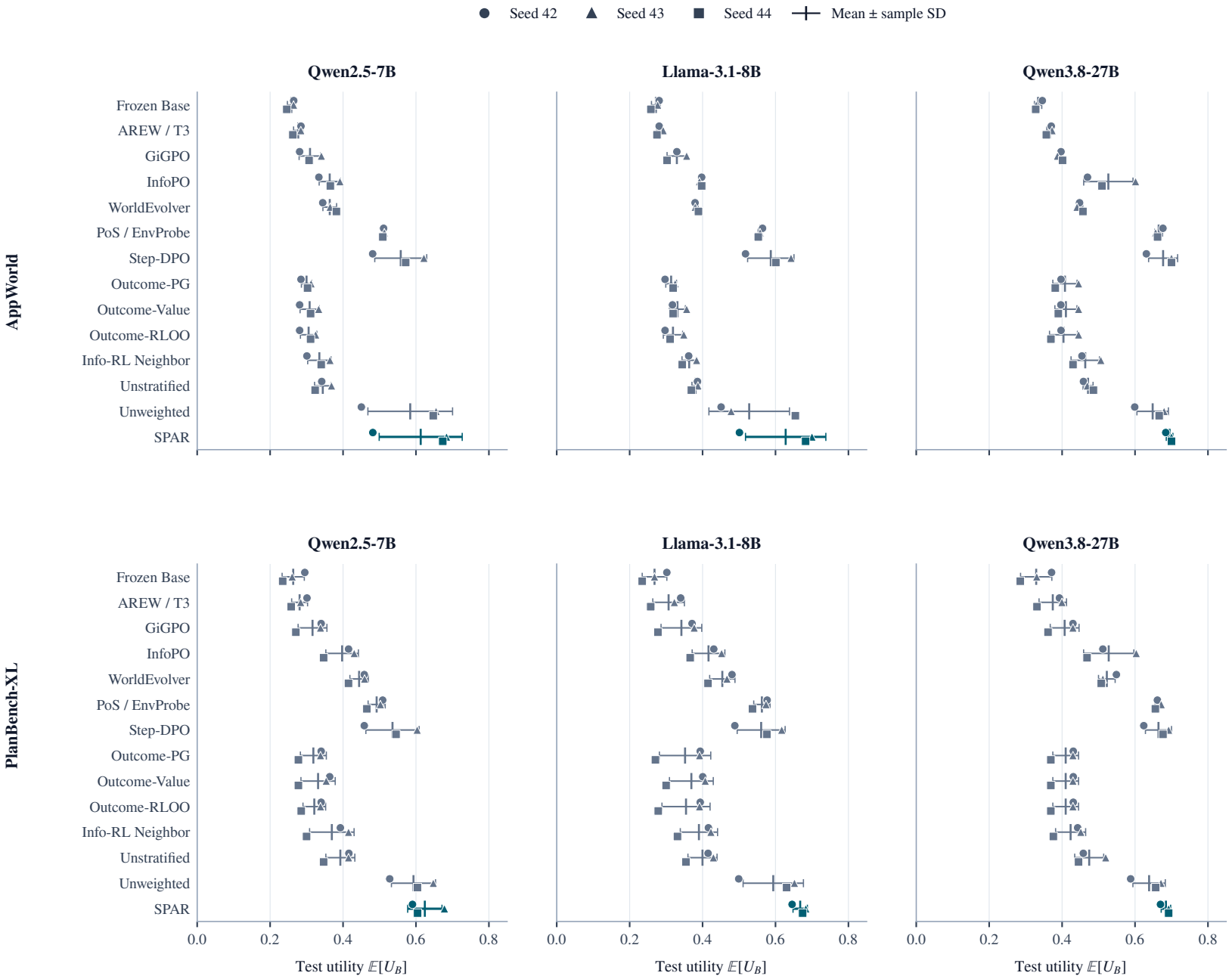
Mean $\mathbb{E}[U_B]$. Test split; mean across seeds 42, 43, 44. AppWorld: 168 tasks; PlanBench-XL: 90 queries.

RQ1 | Test success across all methods

	AppWorld			PlanBench-XL		
Method	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B	Qwen2.5-7B	Llama-3.1-8B	Qwen3.8-27B
Frozen Base	34.9	36.9	46.0	35.9	36.7	45.6
AREW / T3	37.7	38.5	50.8	38.5	42.2	52.2
GiGPO	42.5	45.4	55.2	43.7	47.4	57.0
InfoPO	50.4	55.2	74.6	55.6	58.5	74.8
WorldEvolver	50.4	53.4	63.1	62.6	64.1	74.1
PoS / EnvProbe	72.4	80.6	97.2	70.0	81.9	97.0
Step-DPO	79.4	83.5	96.6	75.9	79.6	94.8
Outcome-PG	41.1	43.1	56.9	44.1	48.9	57.4
Outcome-Value	42.3	45.6	57.3	45.9	51.5	57.4
Outcome-RLOO	41.9	43.8	56.3	44.4	49.3	57.4
Info-RL Neighbor	46.2	50.4	65.1	51.5	54.4	59.6
Unstratified	47.6	53.0	66.3	54.8	55.9	67.0
Unweighted	88.5	77.8	96.4	90.4	90.0	94.8
SPAR	**88.5**	**90.3**	**99.6**	**91.5**	**98.1**	**97.8**

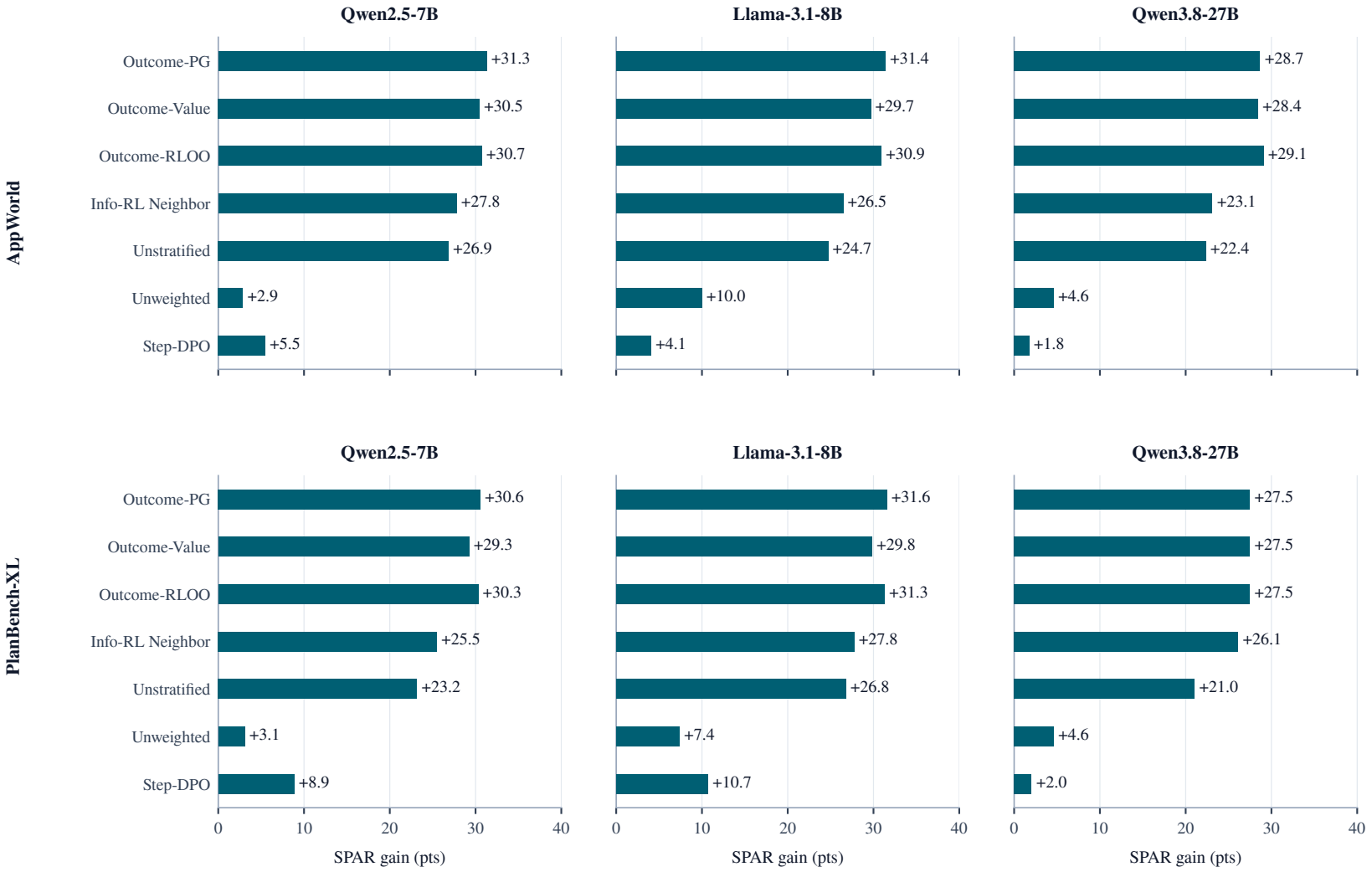
Mean success rate (%). Test split; mean across seeds 42, 43, 44. AppWorld: 168 tasks; PlanBench-XL: 90 queries.

RQ1 | Individual seeds and test-utility variation



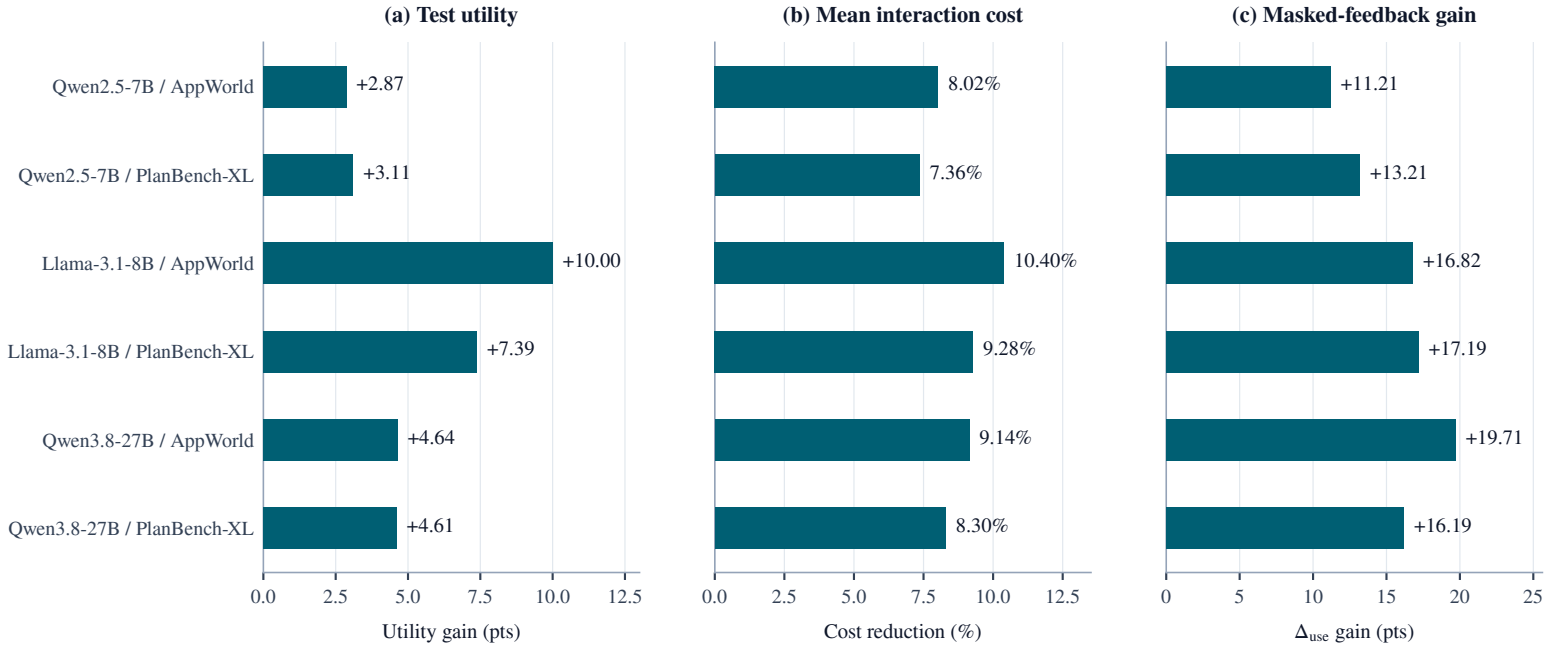
All 252 seed-level results. Points show individual seeds; horizontal intervals show the stored three-seed sample SD. SPAR is highlighted.

RQ2 | Utility gains over mechanism controls



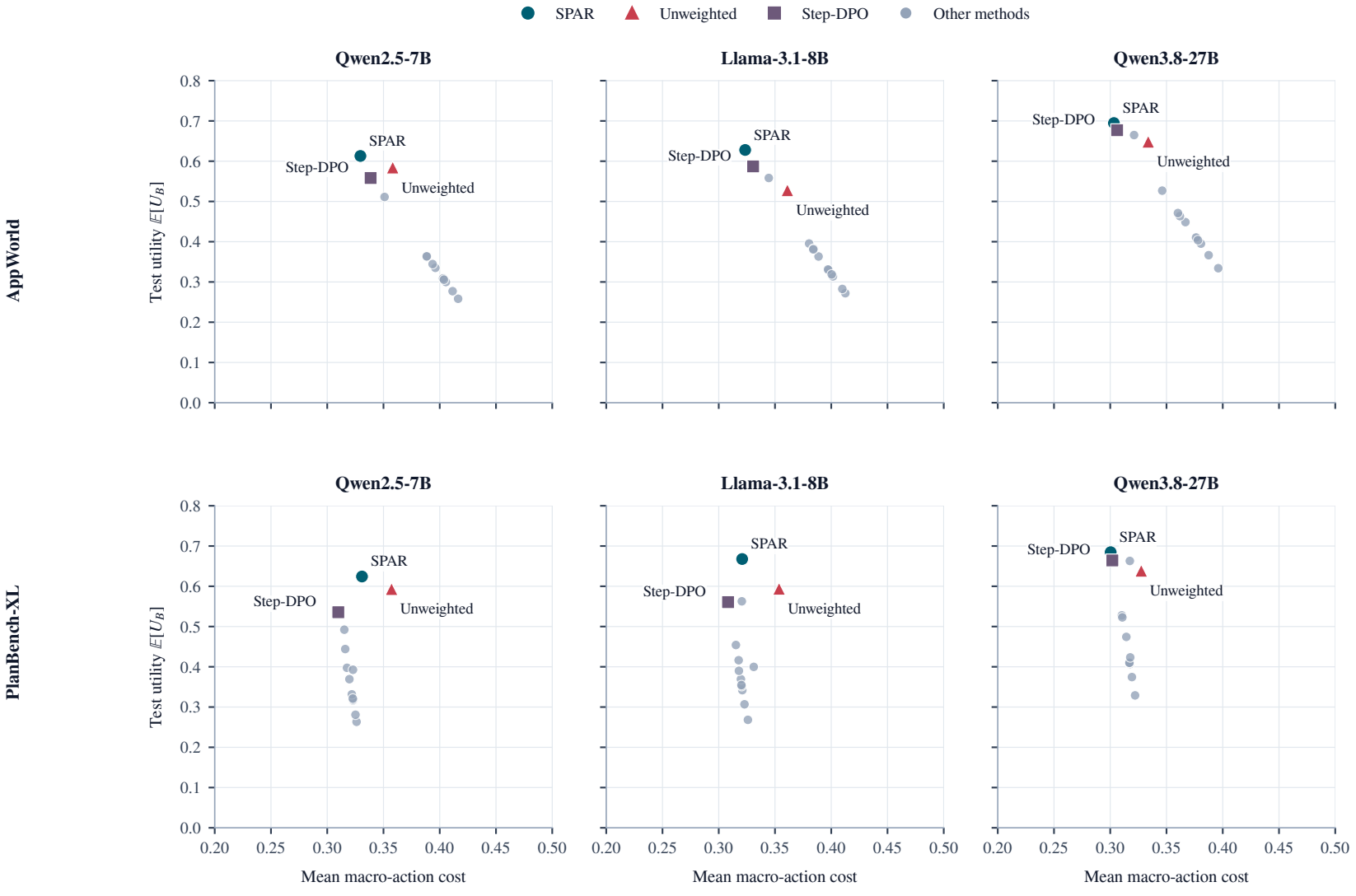
Each bar is 100 × (SPAR mean test utility − reference mean test utility). All contrasts use the same saved model and benchmark setting.

RQ2 | Aligned weighting versus unweighted branching



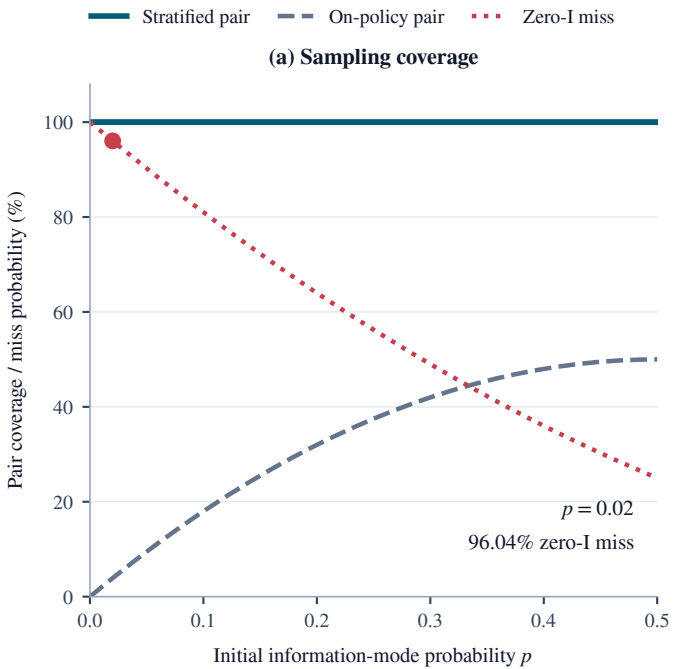
Contrasts of saved means. Cost reduction uses Unweighted as the denominator; feedback gain uses the fixed 24-task development diagnostic.

RQ2 | Test utility and interaction cost



Each point is one method's three-seed mean. The six panels share axes; higher utility and lower macro-action cost are preferred.

RQ3 | Sampling coverage and gradient agreement



(b) Gradient agreement

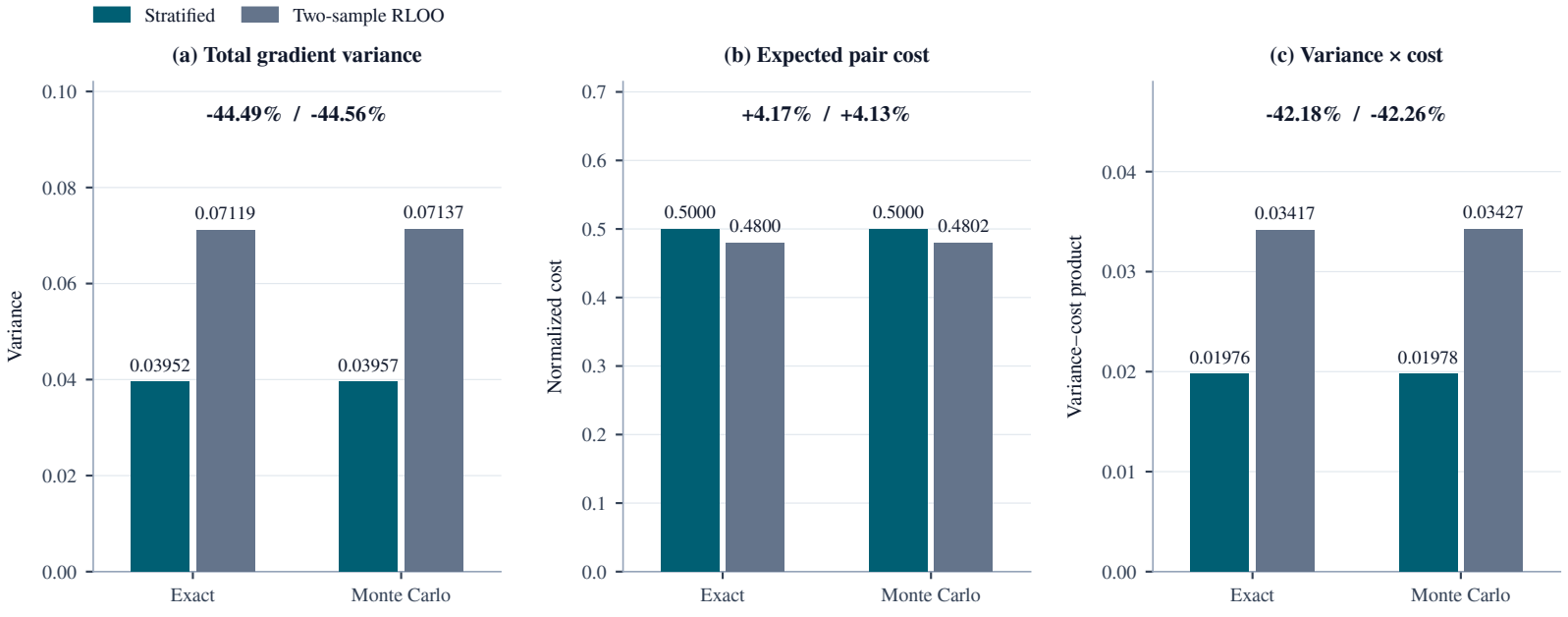
Exact objective: 58/125 = 0.4640

Estimator	g_1	g_2	g_3
Analytical	+0.03840	+0.04480	+0.00000
Stratified exact	+0.03840	+0.04480	+0.00000
RLOO exact	+0.03840	+0.04480	+0.00000
Autograd	+0.03840	+0.04480	+0.00000
Stratified MC	+0.03814	+0.04439	-0.00031
RLOO MC	+0.03736	+0.04529	-0.00164

Variance identity: LHS = RHS = 0.0164736

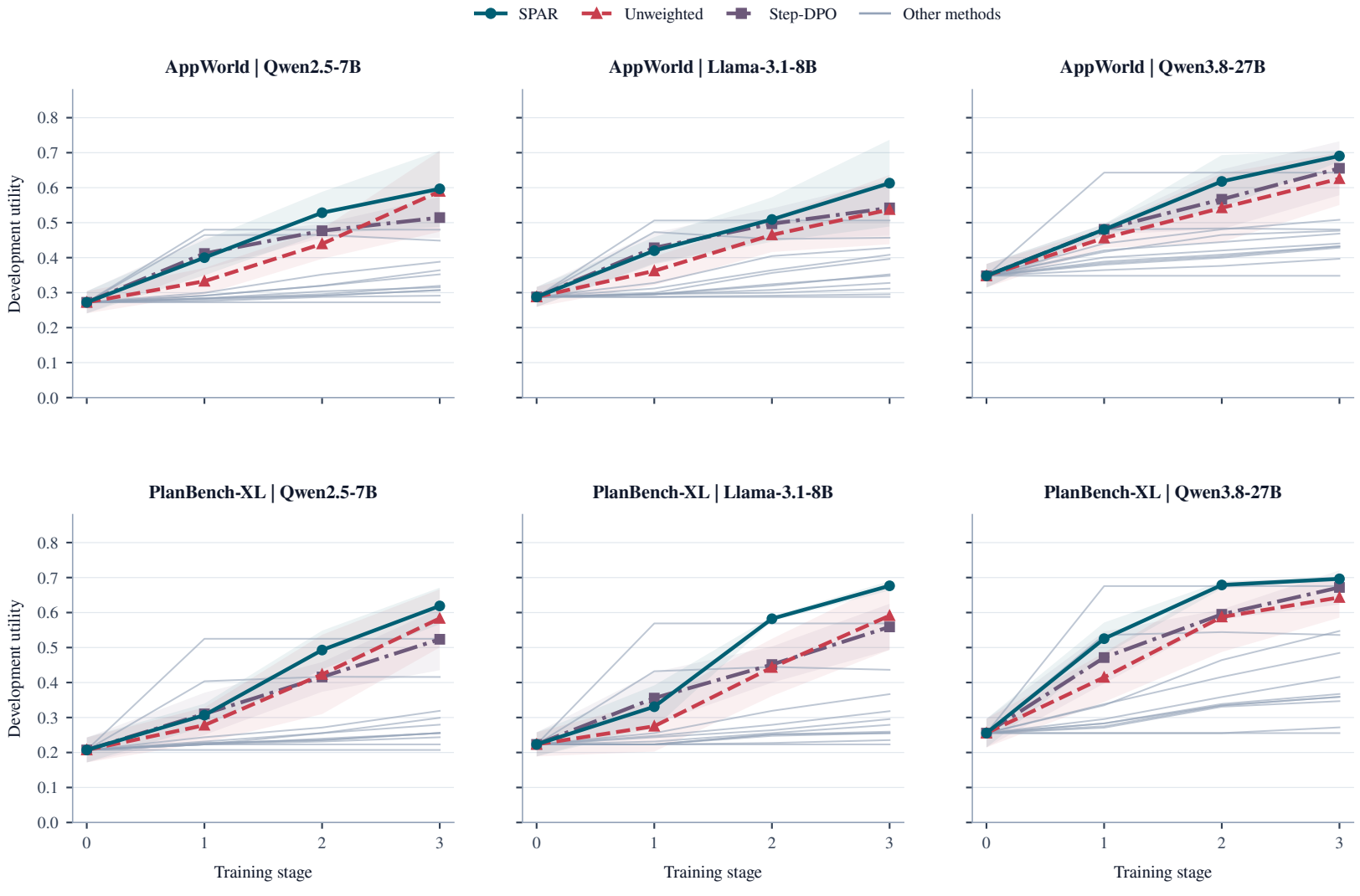
Exact two-stage setting. MC: 10,000 episodes. Gradient coordinates follow the stored parameter-vector order.

RQ3 | Exact and Monte Carlo variance–cost comparison



Percentages compare stratified with RLOO (Exact / Monte Carlo); MC uses 10,000 episodes.

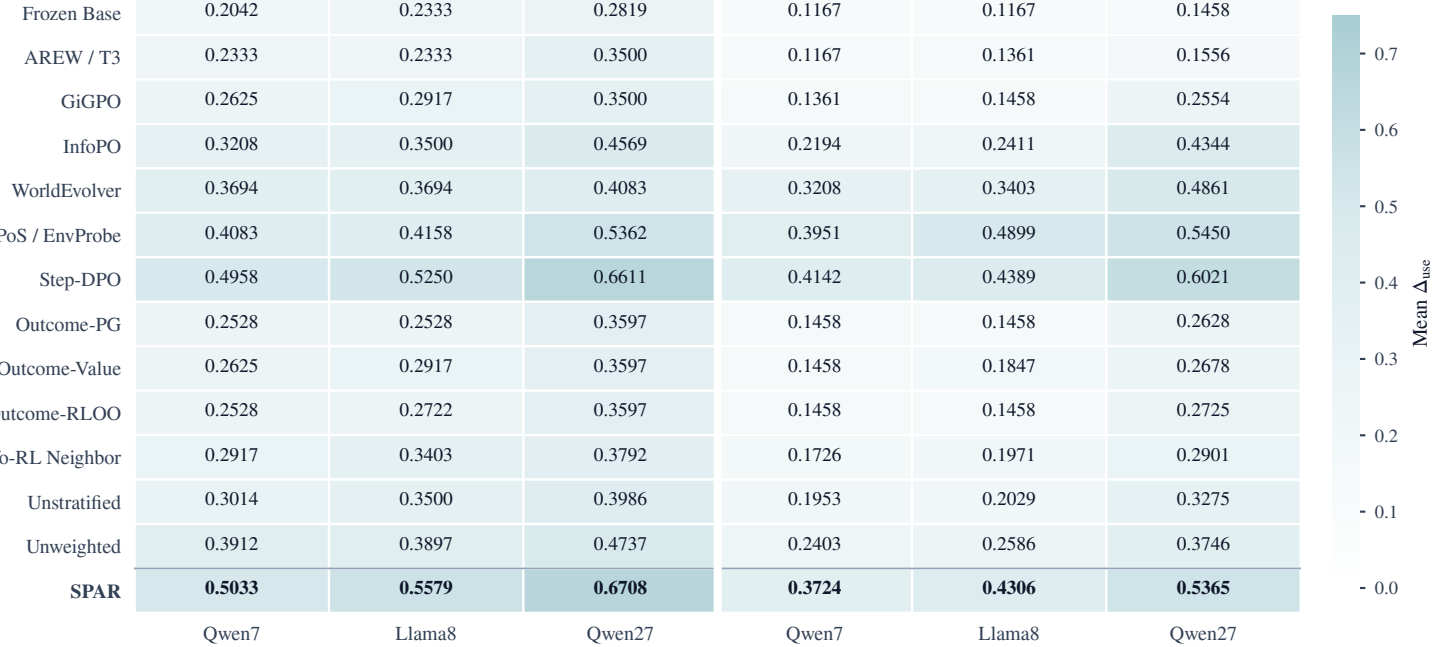
RQ4 | Utility across training stages

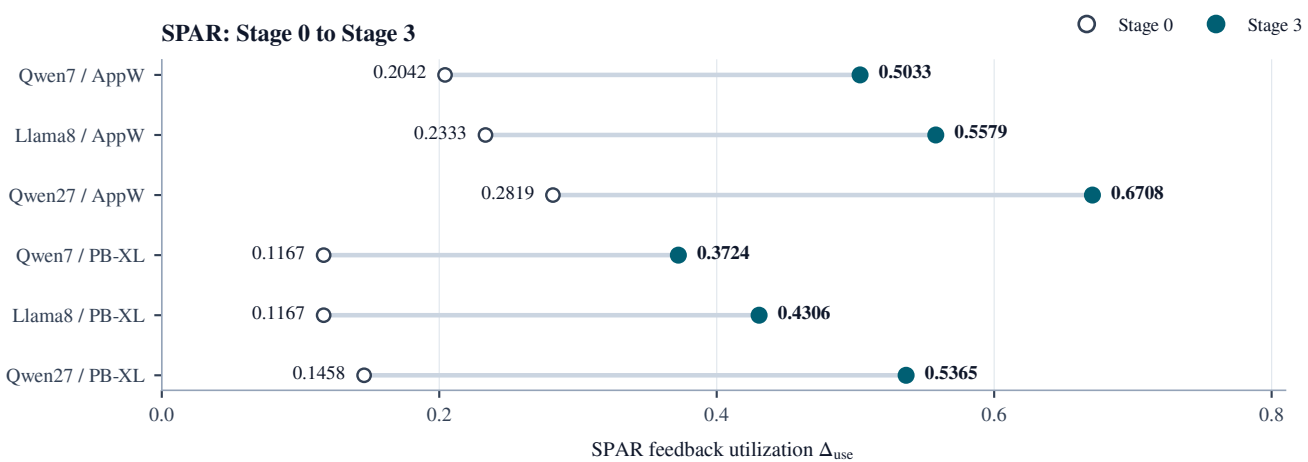


All 14 methods; curves are three-seed means. Bands show sample SD for SPAR, Unweighted and Step-DPO.

RQ4 | Masked-feedback utilization

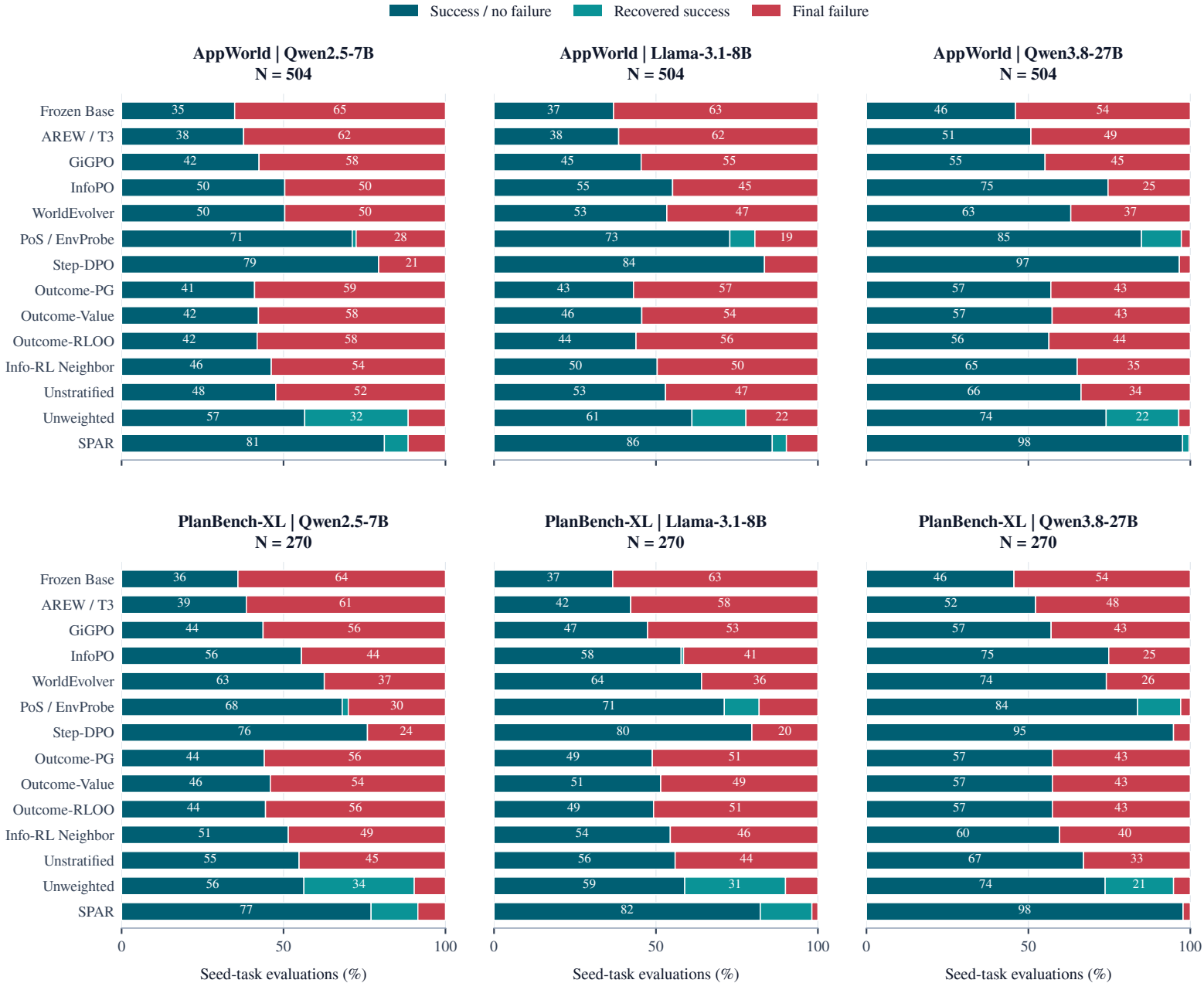
	AppWorld			PlanBench-XL		
	Qwen7	Llama8	Qwen27	Qwen7	Llama8	Qwen27
Frozen Base	0.2042	0.2333	0.2819	0.1167	0.1167	0.1458
AREW / T3	0.2333	0.2333	0.3500	0.1167	0.1361	0.1556
GiGPO	0.2625	0.2917	0.3500	0.1361	0.1458	0.2554
InfoPO	0.3208	0.3500	0.4569	0.2194	0.2411	0.4344
WorldEvolver	0.3694	0.3694	0.4083	0.3208	0.3403	0.4861
PoS / EnvProbe	0.4083	0.4158	0.5362	0.3951	0.4899	0.5450
Step-DPO	0.4958	0.5250	0.6611	0.4142	0.4389	0.6021
Outcome-PG	0.2528	0.2528	0.3597	0.1458	0.1458	0.2628
Outcome-Value	0.2625	0.2917	0.3597	0.1458	0.1847	0.2678
Outcome-RLOO	0.2528	0.2722	0.3597	0.1458	0.1458	0.2725
Info-RL Neighbor	0.2917	0.3403	0.3792	0.1726	0.1971	0.2901
Unstratified	0.3014	0.3500	0.3986	0.1953	0.2029	0.3275
Unweighted	0.3912	0.3897	0.4737	0.2403	0.2586	0.3746
SPAR	**0.5033**	**0.5579**	**0.6708**	**0.3724**	**0.4306**	**0.5365**





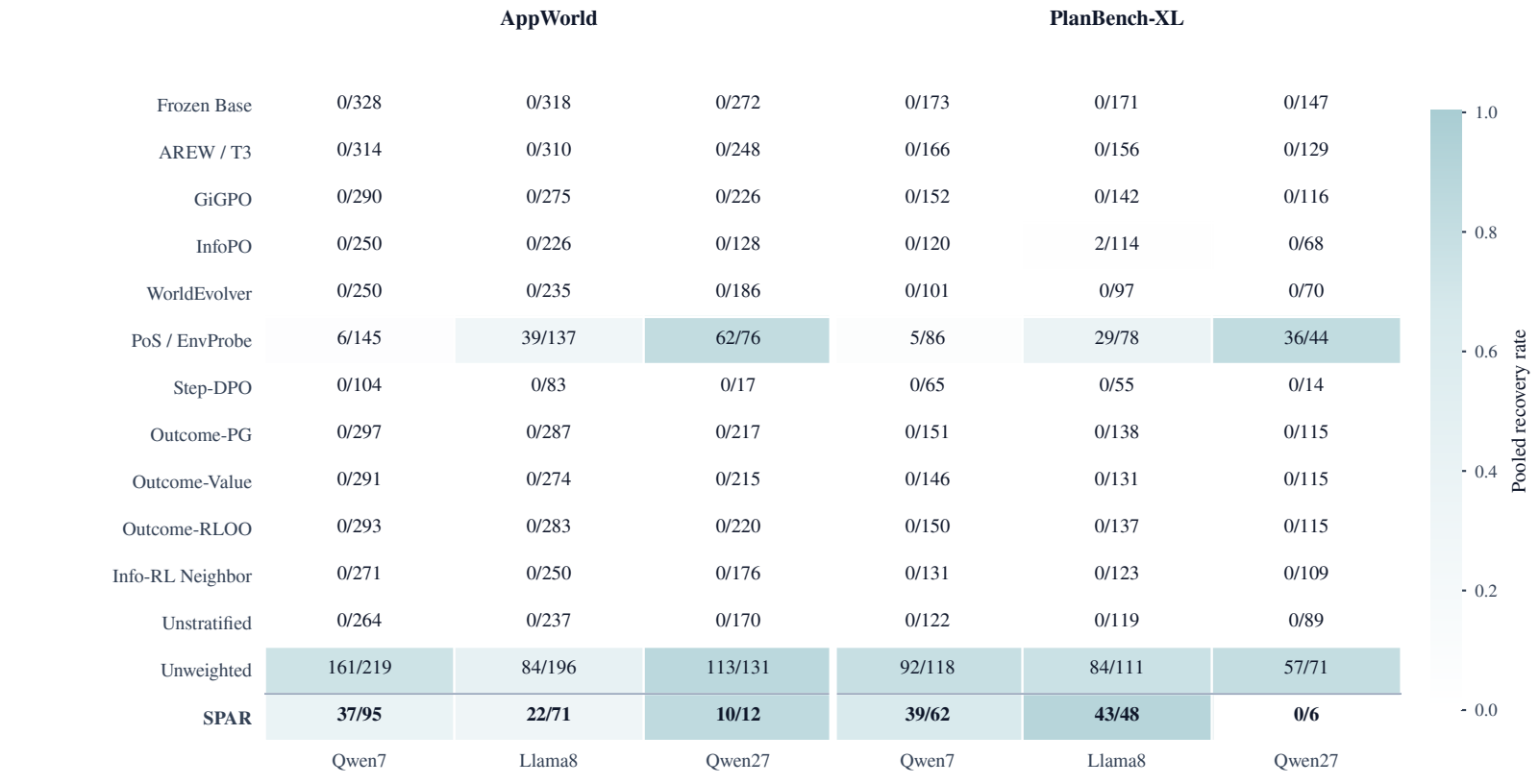
Fixed 24-task development diagnostic; first information feedback is masked. Top: three-seed means. Bottom: SPAR stage endpoints.

RQ4 | Trajectory outcomes across all methods



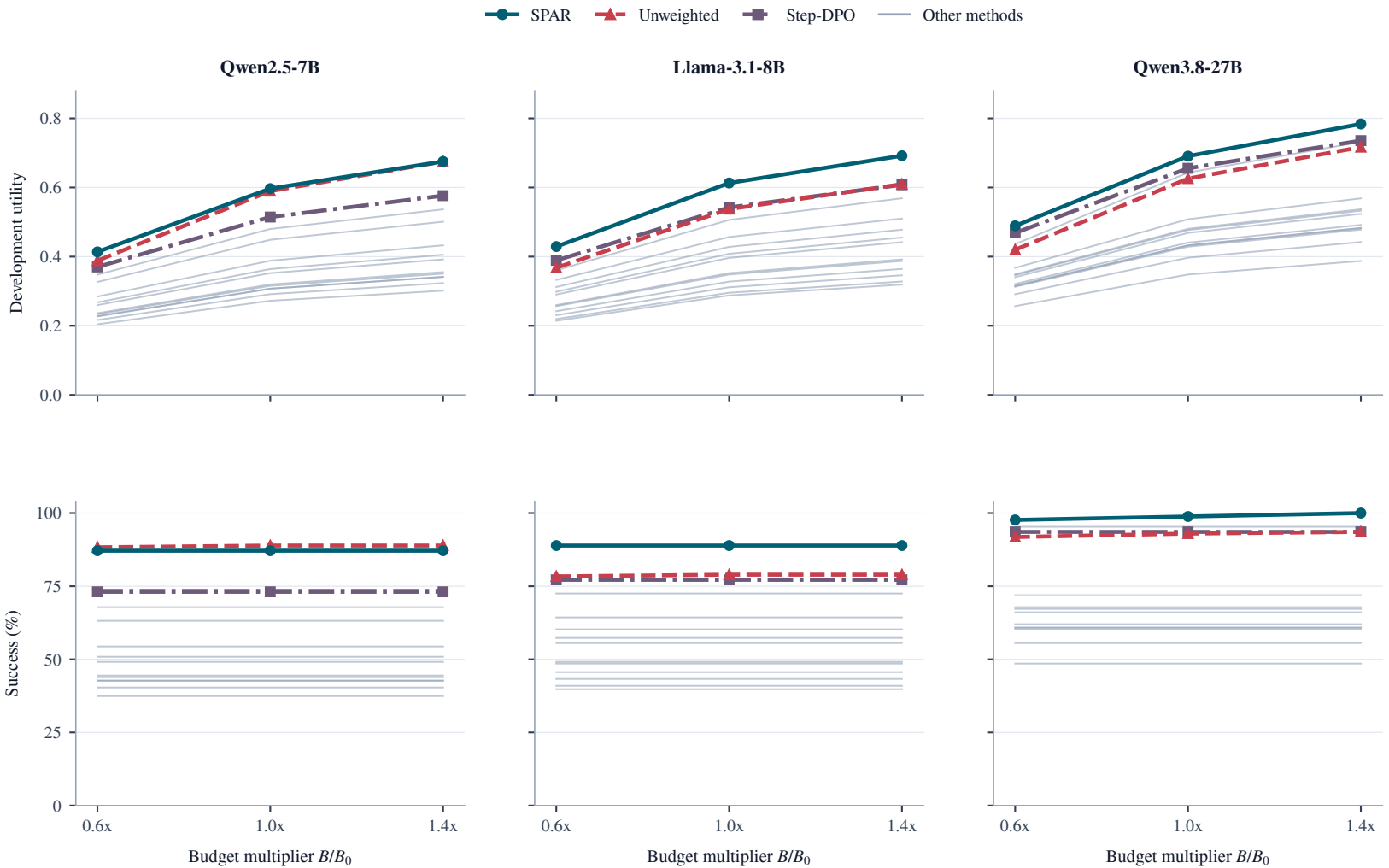
Counts pooled over three seeds. Success / no failure means success without a recorded intermediate failure; recovered success follows a failure.

RQ4 | Recovery conditional on an intermediate failure



Each cell is recovered successes / evaluations with a recorded failure, pooled across seeds. Color encodes the ratio; dash means a zero denominator.

RQ5 | Utility and success across available budgets

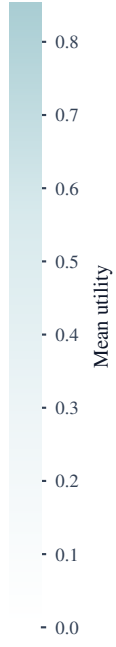


AppWorld development set; all 14 methods. Values are three-seed means from the saved budget sweep.

RQ5 | Complete budget utility matrix

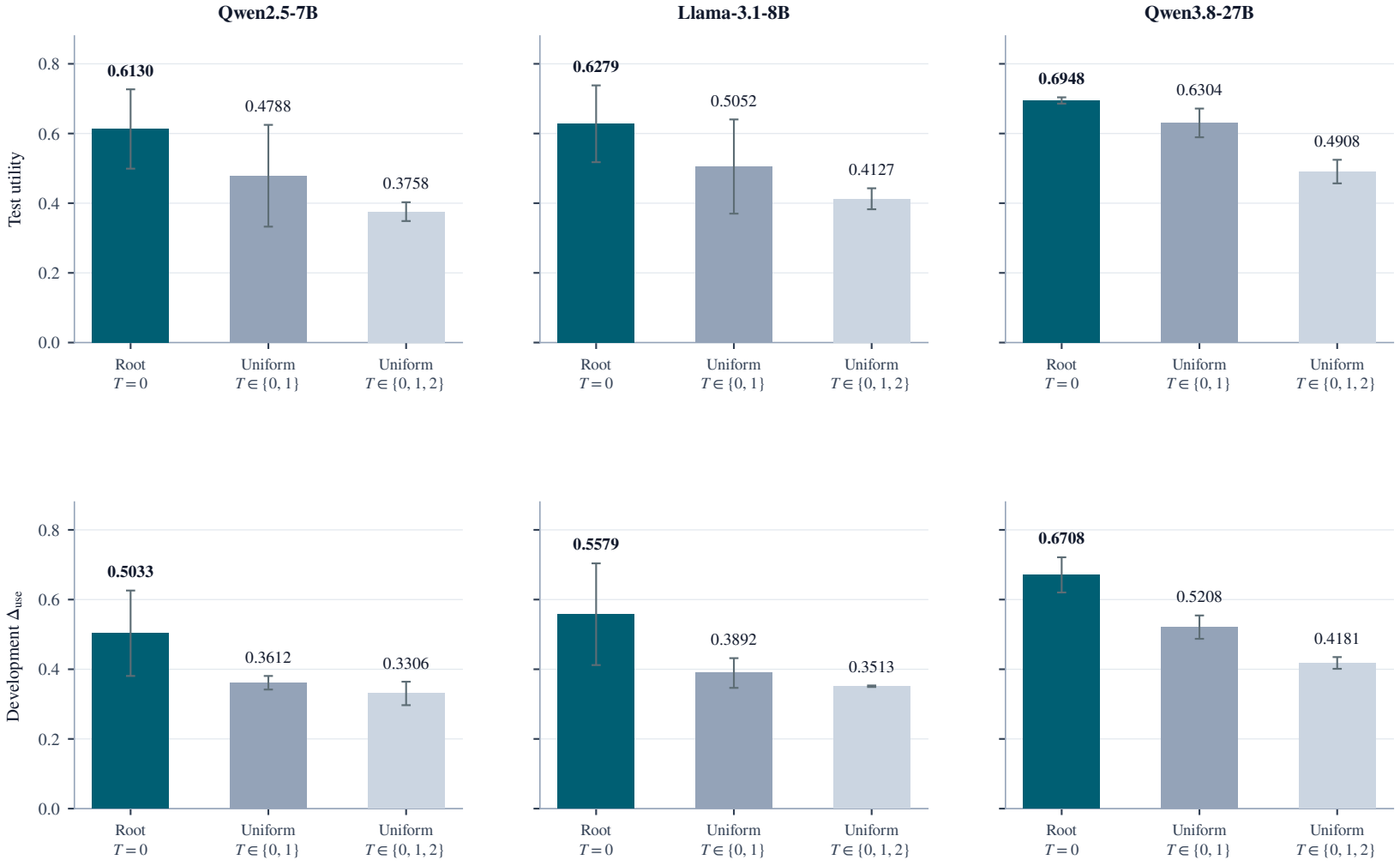
	Qwen2.5-7B			Llama-3.1-8B			Qwen3.8-27B		
Frozen Base	0.2043	0.2723	0.3014	0.2144	0.2877	0.3191	0.2567	0.3482	0.3874
AREW / T3	0.2166	0.2913	0.3234	0.2195	0.2954	0.3280	0.2910	0.3968	0.4422
GiGPO	0.2333	0.3154	0.3506	0.2567	0.3482	0.3874	0.3150	0.4323	0.4825
InfoPO	0.2844	0.3882	0.4327	0.3121	0.4282	0.4779	0.3674	0.5082	0.5685
WorldEvolver	0.3267	0.4487	0.5009	0.3326	0.4568	0.5101	0.3485	0.4805	0.5370
PoS / EnvProbe	0.3478	0.4800	0.5367	0.3602	0.5062	0.5688	0.4363	0.6430	0.7317
Step-DPO	0.3702	0.5145	0.5764	0.3891	0.5422	0.6079	0.4686	0.6554	0.7355
Outcome-PG	0.2275	0.3073	0.3414	0.2304	0.3113	0.3460	0.3158	0.4327	0.4829
Outcome-Value	0.2363	0.3195	0.3552	0.2596	0.3523	0.3920	0.3209	0.4405	0.4917
Outcome-RLOO	0.2275	0.3073	0.3414	0.2421	0.3277	0.3644	0.3129	0.4287	0.4783
Info-RL Neighbor	0.2596	0.3523	0.3920	0.2903	0.3964	0.4419	0.3405	0.4687	0.5236
Unstratified	0.2676	0.3641	0.4054	0.2982	0.4082	0.4553	0.3464	0.4768	0.5327
Unweighted	0.3883	0.5895	0.6750	0.3678	0.5374	0.6094	0.4199	0.6257	0.7166
SPAR	**0.4136**	**0.5967**	**0.6752**	**0.4290**	**0.6130**	**0.6918**	**0.4891**	**0.6906**	**0.7837**
	0.6x	1.0x	1.4x	0.6x	1.0x	1.4x	0.6x	1.0x	1.4x

Budget multiplier B/B_0

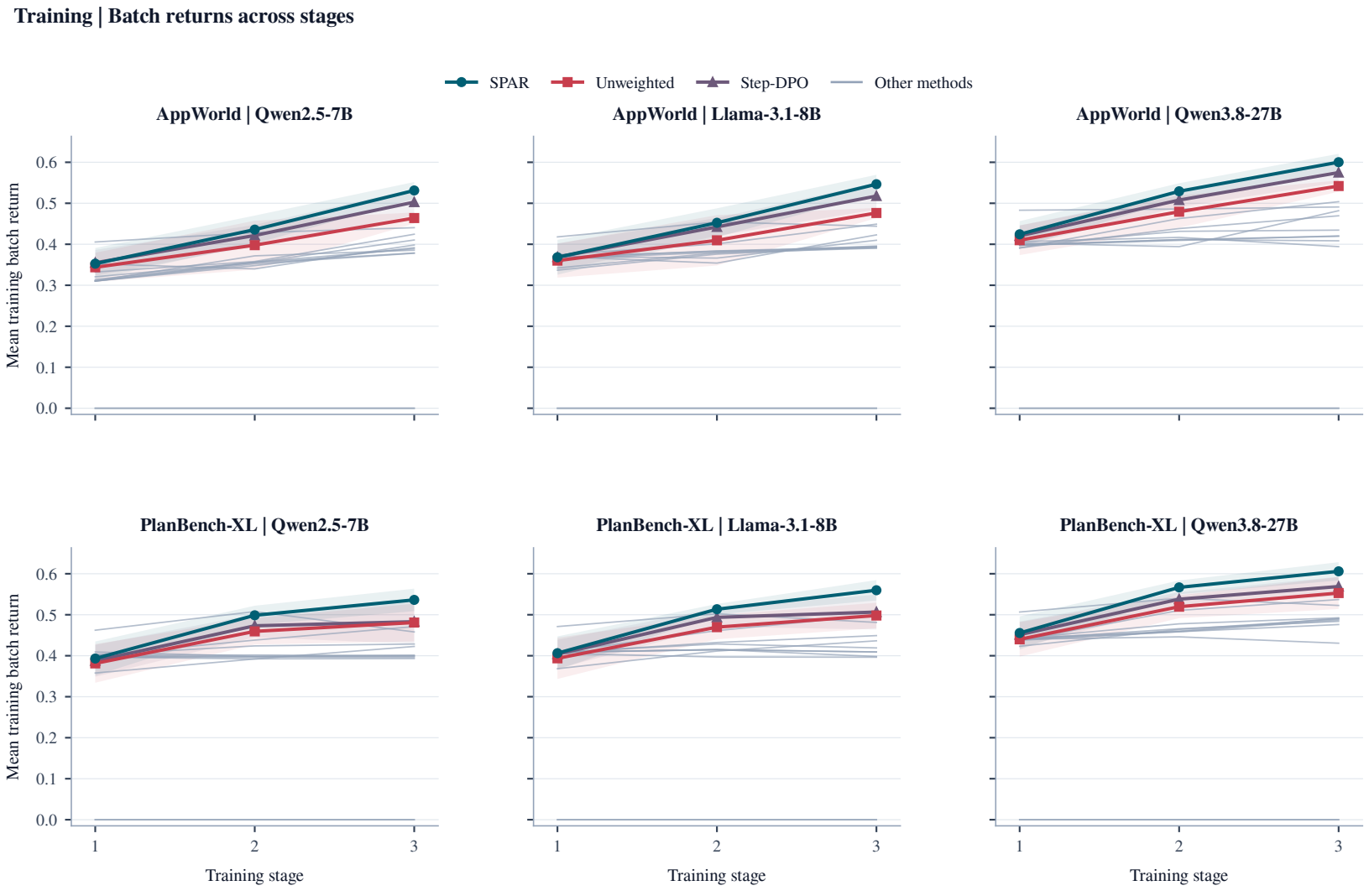


AppWorld development set; 3 backbones x 14 methods x 3 budgets. Cells are three-seed mean utility.

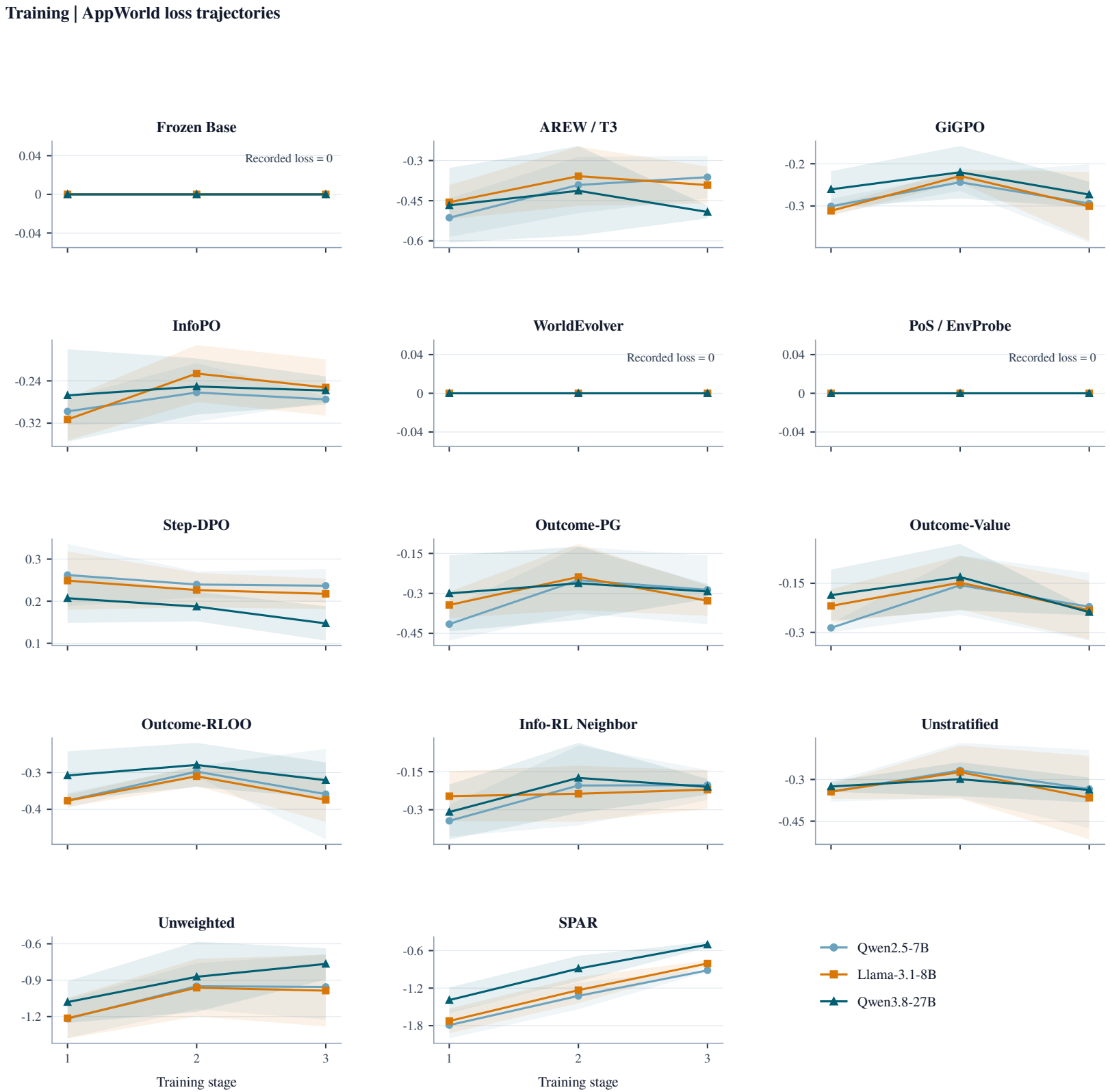
RQ5 | Where to split the training trajectory



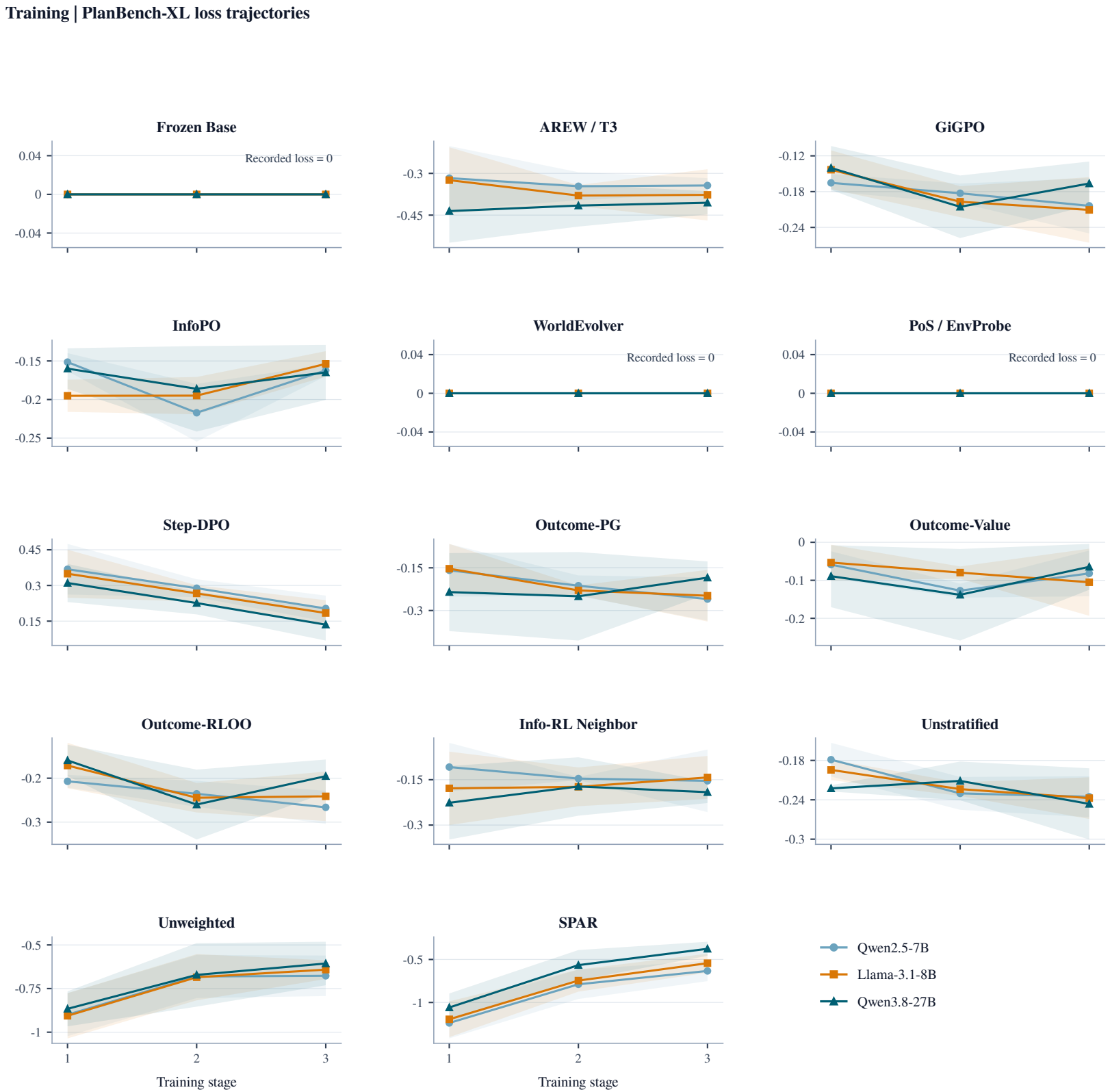
AppWorld; bar heights are three-seed means with sample SD. Uniform schedules sample the split location T from the displayed set.



Lines are three-seed means; bands show sample SD for highlighted methods. Recorded zero-update values are retained.



Points and bands show mean and sample SD across seeds 42, 43, 44. Each method uses its own loss scale.



Points and bands show mean and sample SD across seeds 42, 43, 44. Each method uses its own loss scale.

Resources | Interaction and update counts

Environment steps

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0.0	0.0	0.0	0.0	0.0	0.0
AREW / T3	1434.0	1407.7	1405.7	1289.7	1307.0	1280.0
GiGPO	1428.7	1401.3	1397.7	1294.7	1299.7	1276.0
InfoPO	1426.0	1408.0	1371.0	1288.0	1269.3	1265.7
WorldEvolver	1319.7	1309.3	1306.3	1196.3	1193.3	1204.0
PoS / EnvProbe	0.0	0.0	0.0	0.0	0.0	0.0
Step-DPO	1362.0	1365.0	1366.3	1303.3	1305.0	1309.7
Outcome-PG	1438.0	1402.7	1407.7	1306.3	1302.3	1271.0
Outcome-Value	1434.3	1396.7	1387.0	1298.3	1301.7	1269.7
Outcome-RLOO	1433.3	1397.3	1406.7	1304.3	1303.3	1271.0
Info-RL Neighbor	1436.3	1417.0	1392.3	1297.3	1287.3	1268.3
Unstratified	1435.7	1422.3	1381.0	1292.3	1292.3	1274.3
Unweighted	1372.0	1369.3	1368.7	1313.7	1314.0	1330.3
SPAR	**1367.3**	**1360.7**	**1366.7**	**1322.3**	**1323.7**	**1329.0**

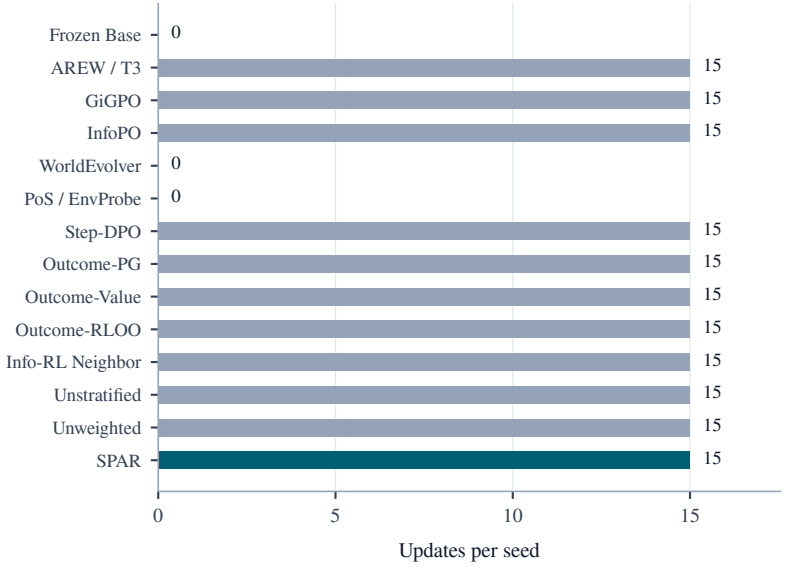
Steps: 0, 200, 400, 600, 800, 1000, 1200, 1400

Accounted tokens (k)

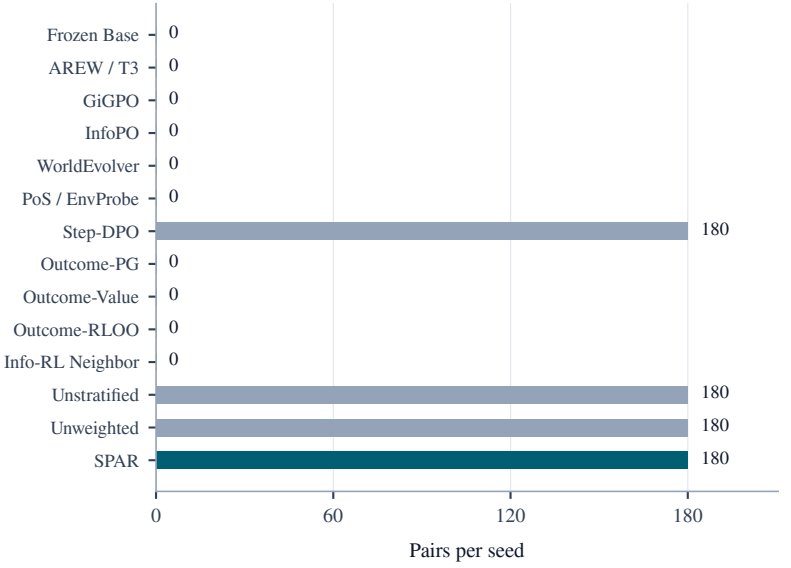
	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0.0	0.0	0.0	0.0	0.0	0.0
AREW / T3	489.0	471.2	479.7	582.5	570.3	579.1
GiGPO	487.1	468.2	475.3	586.1	566.4	576.4
InfoPO	488.0	471.4	460.9	581.6	545.9	571.0
WorldEvolver	582.9	573.1	579.8	655.2	632.5	663.7
PoS / EnvProbe	0.0	0.0	0.0	0.0	0.0	0.0
Step-DPO	448.4	445.1	453.9	593.7	571.3	603.2
Outcome-PG	491.0	469.2	480.9	594.5	568.4	572.8
Outcome-Value	490.1	466.0	469.9	589.8	567.8	572.2
Outcome-RLOO	488.6	465.0	480.6	593.3	568.6	572.8
Info-RL Neighbor	493.5	476.1	471.8	589.3	558.1	571.9
Unstratified	489.9	476.8	462.1	584.5	561.1	578.2
Unweighted	454.0	446.4	454.5	601.2	578.0	619.3
SPAR	**450.5**	**441.5**	**453.6**	**607.3**	**584.5**	**618.3**

k tokens: 0, 100, 200, 300, 400, 500, 600

Gradient updates



Snapshot pairs

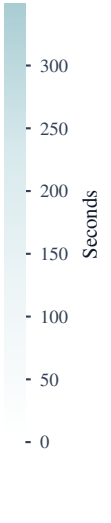


Per-seed means (3 seeds). Lower panels: counts shared by all six settings. Q7/L8/Q27: Qwen2.5-7B/Llama-3.1-8B/Qwen3.8-27B; AW/PB: AppWorld/PlanBench-XL.

Resources | Recorded training time

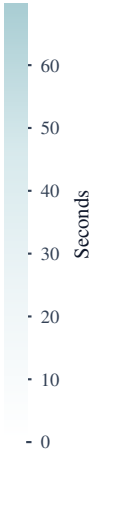
Amortized model time (s)

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0	0	0	0	0	0
AREW / T3	150.2	89.4	277.1	82.7	113.6	322.4
GiGPO	149.0	89.0	274.2	83.2	112.5	320.5
InfoPO	147.9	90.6	265.7	82.9	109.5	317.0
WorldEvolver	124.3	81.5	245.4	75.0	99.5	289.3
PoS / EnvProbe	0	0	0	0	0	0
Step-DPO	124.4	85.1	261.1	85.1	115.7	337.4
Outcome-PG	152.0	89.0	277.9	84.4	112.9	318.2
Outcome-Value	150.3	88.6	271.0	83.9	113.2	317.8
Outcome-RLOO	150.5	88.4	277.8	84.2	112.9	318.3
Info-RL Neighbor	150.8	90.4	272.3	83.9	111.2	318.2
Unstratified	149.6	90.1	266.9	83.0	113.4	322.0
Unweighted	125.8	85.5	261.8	86.3	117.2	347.1
SPAR	**125.1**	**84.5**	**261.1**	**87.1**	**118.6**	**346.5**



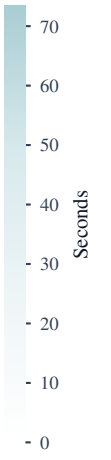
Gradient-update time (s)

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0	0	0	0	0	0
AREW / T3	26.84	26.54	4.40	2.58	2.76	2.46
GiGPO	27.66	65.24	4.41	2.56	2.70	2.48
InfoPO	29.41	56.07	5.23	3.29	3.47	3.15
WorldEvolver	0	0	0	0	0	0
PoS / EnvProbe	0	0	0	0	0	0
Step-DPO	35.91	21.75	6.17	4.35	4.63	4.24
Outcome-PG	22.52	35.16	5.26	3.29	3.55	3.22
Outcome-Value	31.89	69.34	3.81	2.72	2.88	2.54
Outcome-RLOO	31.49	35.66	3.81	3.28	3.51	3.12
Info-RL Neighbor	26.66	41.48	3.70	3.41	3.64	3.10
Unstratified	11.00	25.25	2.52	2.30	2.23	2.07
Unweighted	2.39	2.50	2.37	2.21	2.20	2.18
SPAR	**24.68**	**54.84**	**4.14**	**2.24**	**2.36**	**2.17**



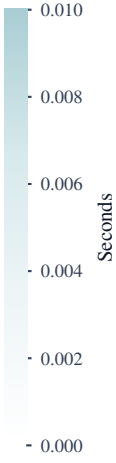
Wall-clock time (s)

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0	<0.01	0	0	0	0
AREW / T3	30.65	30.52	7.88	5.73	6.12	5.42
GiGPO	31.51	68.98	7.92	5.70	5.96	5.51
InfoPO	33.26	60.01	8.95	6.49	6.96	6.31
WorldEvolver	3.93	3.98	3.63	3.49	3.73	3.45
PoS / EnvProbe	0.59	0.65	0.60	0.54	0.68	0.59
Step-DPO	40.13	26.03	10.12	7.97	8.52	7.85
Outcome-PG	26.32	38.96	8.79	6.38	6.98	6.25
Outcome-Value	35.87	73.27	7.38	5.85	6.35	5.48
Outcome-RLOO	35.42	39.54	7.48	6.41	7.00	6.07
Info-RL Neighbor	30.50	45.34	7.30	6.60	7.14	6.08
Unstratified	15.04	29.46	6.45	5.89	5.84	5.43
Unweighted	6.50	6.85	6.54	6.01	6.01	5.86
SPAR	**28.81**	**59.16**	**8.20**	**6.12**	**6.44**	**6.04**



Snapshot-restore time (s)

	Q7 AW	L8 AW	Q27 AW	Q7 PB	L8 PB	Q27 PB
Frozen Base	0	0	0	0	0	0
AREW / T3	0	0	0	0	0	0
GiGPO	0	0	0	0	0	0
InfoPO	0	0	0	0	0	0
WorldEvolver	0	0	0	0	0	0
PoS / EnvProbe	0	0	0	0	0	0
Step-DPO	0.007	0.003	0.010	0	0.003	0
Outcome-PG	0	0	0	0	0	0
Outcome-Value	0	0	0	0	0	0
Outcome-RLOO	0	0	0	0	0	0
Info-RL Neighbor	0	0	0	0	0	0
Unstratified	0.007	0	0.003	0.003	0	0
Unweighted	0.003	0.007	0.003	0.007	0.003	0.003
SPAR	**0**	**0.003**	**0.003**	**0**	**0**	**0.010**



Per-seed means over 3 seeds. Model time includes cached encoding amortization. AW/PB: AppWorld/PlanBench-XL; Q7/L8/Q27: the three backbones.